## Paper 1: Purnomo, Abdullah & Irawati (2011) *"A System Dynamics Approach to Balancing Wood Supply and Demand for Sustaining the Future Industry"*

**Scope:** Indonesian teak furniture supply chain in Jepara district - from forests to furniture to domestic and international markets.

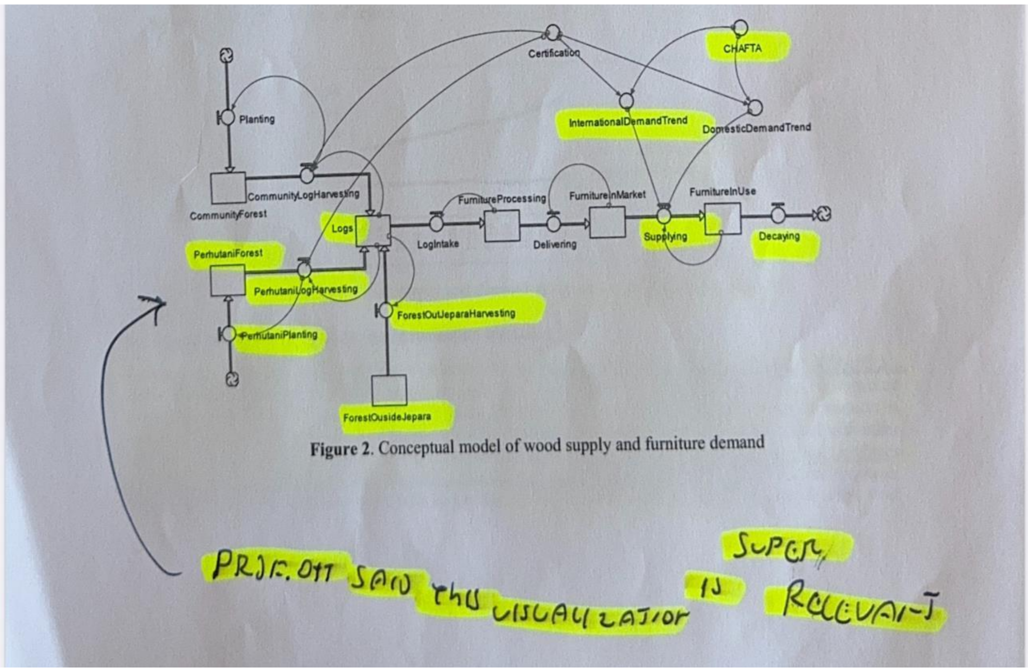
**Method:** System Dynamics, implemented in STELLA.

**Inputs:** Forest area by age class, harvesting rates, wood consumption per workshop (862,056 m3/year total), logging conversion factor (0.7), furniture processing efficiency (30%), domestic and international demand trends.

**Outputs:** Supply vs demand projections (2011-2022), impact of CHAFTA free trade and certification scenarios on the supply-demand balance.

**Model structure:** Covers forest stock (state-owned Perhutani + community), log processing, and furniture market (domestic + international). Key stocks are forest area, logs, and furniture in market.

**Key finding:** Supply cannot keep up with demand under any scenario. CHAFTA and certification reduce demand but don't fix the supply problem. Increasing furniture prices is the only way to incentivise tree growing.



## Paper 2: Ma, Zhang, Gao & Liu (2025) *"Modelling and Evaluating Wood Supply Chain Resilience Factors Based on System Dynamics"* Published in Nature Scientific Reports.

**Scope:** Wood supply chain in Heilongjiang Province, China - from imports to processing to furniture sales and recycling.

**Method:** System Dynamics, implemented in AnyLogic. Validated against historical data (2012-2021).

**Inputs:** GDP of Heilongjiang Province, wood import volumes, port service capacity, logistics infrastructure, production capacity, automation levels, power supply stability, market transactions, recycling levels, number of sales enterprises.

**Outputs:** Long-term resilience simulation across three dimensions showing how different factors interact over time.

**Model structure:** Three subsystems - wood import supply, wood processing & manufacturing, wooden furniture sales & recycling. Contains 8 reinforcing loops, 2 balancing loops, and 5 exogenous links. GDP is the core connecting variable.

**Resilience measured in 3 dimensions:**

Infrastructure (transport, logistics, ports),

Economic (raw material supply, price control, worker skills),

Ecological (sustainable forestry, pollution control, recycling).

**Key finding:** Relying on overseas raw materials makes coordination difficult. Manufacturing needs better efficiency. Balancing domestic and international consumption stabilises the system.

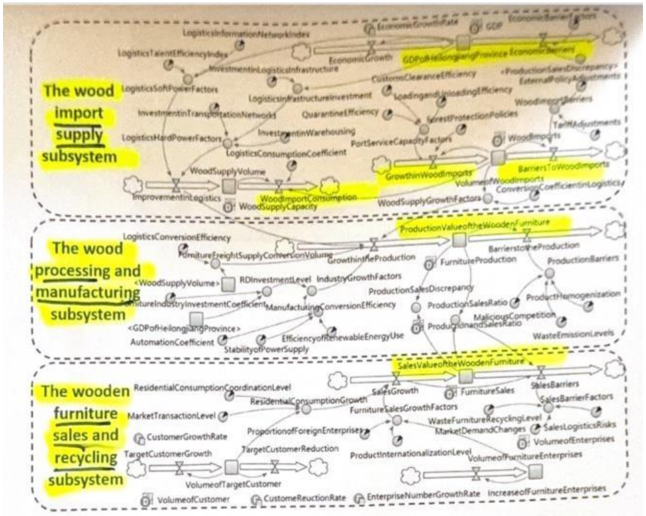



**Fig. 3.** Stock and flow diagram of the wood supply chain system in Heilongjiang Province.

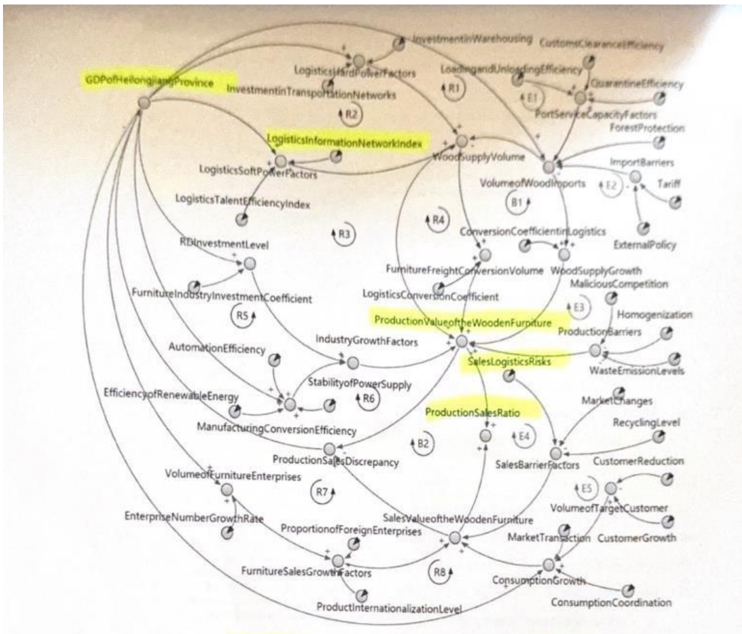



**Fig. 2.** Causality diagram of the wood supply chain system in Heilongjiang Province.

**Paper 3: Svirakova (2018)** *"Close the Loop! System Dynamics Modelling in Service Design"* Published in Systems (MDPI).

- **Scope:** Service design for a university course (Project Management subject) - not a supply chain. Models why students don't register for a course and how to fix it.

- **Method:** System Dynamics, implemented in Vensim. Follows Sterman's 5-step SD process. Uses qualitative data (interviews with students and lecturer) converted into a quantitative model through Grounded Theory coding.

- **Inputs:** Number of students in the subject (initial value: 3), student capacity limitation (15), advantaged time positioning in the study plan (0-100 scale), quality of communication, student's adjustment time.

- **Outputs:** Behavior-over-time graphs showing number of students, lecturer's motivation, and confidence in subject quality across 12 years under different scenarios.

- **Model structure:** 3 stocks (number of students, confidence in subject quality, lecturer's motivation), 3 flows, 2 reinforcing loops (R1, R2), 1 balancing loop (B). Full equations provided in the appendix.

- **3 scenarios tested:** (1) No advantage in study plan position - system stays flat at 3 students. (2) High advantage position - students reach 15 by year 4 but then crash due to balancing loop. (3) Gradual positioning change - stable growth.

- **Key finding:** The feedback loop between study plan positioning, student numbers, lecturer motivation, and content quality creates a vicious or virtuous cycle. Small policy changes (moving the course earlier in the study plan) can dramatically change outcomes.

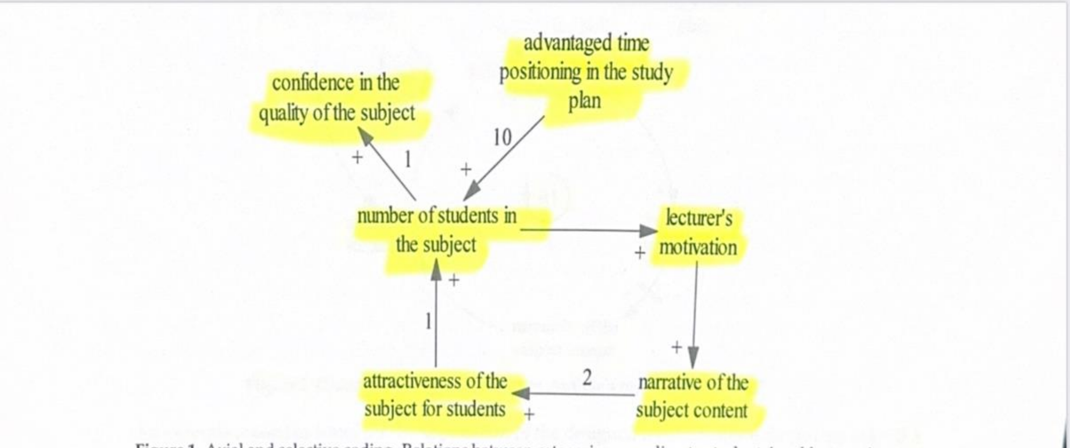



**Figure 1.** Axial and selective coding. Relations between categories according to students' and lecturer's opinions. Source: Author's own diagram, 2017.

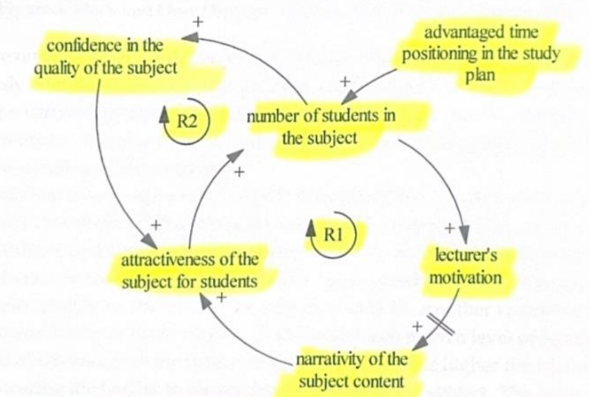



**Figure 2.** Casual Loop Diagram. Source: Author's own diagram, 2017.

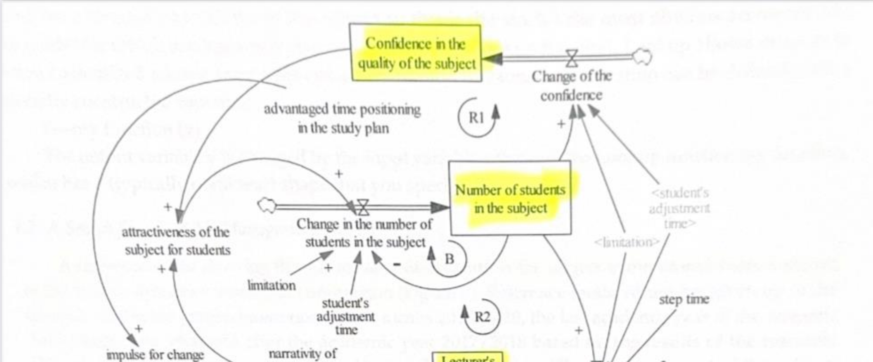



**Figure 3.** Stock and Flow Diagram. Source: Author's own diagram, 2018.

**Paper 4: Bashiri, Tjahjono, Lazell, Ferreira & Perdana (2021)** *"The Dynamics of Sustainability Risks in the Global Coffee Supply Chain: A Case of Indonesia-UK"* Published in Sustainability (MDPI), Vol. 13, No. 2, 589.

- **Scope:** Indonesia-UK coffee supply chain - from farming to collection to processing to export (domestic, UK, and other countries).

- **Method:** System Dynamics, built in Vensim. Data extracted from stakeholder interviews (farmers + exporters in Indonesia, importers in UK) and validated against literature. Also combined with TOPSIS multi-criteria decision making to rank scenarios.

- **Inputs:** 5 stocks (uncollected coffee, collected coffee, domestic usage, UK exported coffee, other countries' exported coffee), 10 dynamic variables (climate change, coffee price, agricultural productivity, labor issues, land use, consumption pattern, quality of coffee, labor supply, timeliness of supply, total exported coffee), 14 parameters (exchange rate, trade policies, transportation disruptions, area increasing rate, etc.).

- **Outputs:** Supply chain behavior over 1-year and 10-year simulated run times under 5 different what-if scenarios. Total collected coffee validated at 940,397 kg after year 1 - close to expected value.

- **Model structure:** Built in 3 stages - Farmers' view model, Exporters' view model, then integrated into one model. 4 flows connecting the stocks. Cause-and-effect relationships mapped before building the SFD.

- **Validation:** Walk-through method, cause-and-effect table consistency check, sensitivity analysis (tested area increasing rate, climate change factor, transportation disruption parameters).

- **5 scenarios tested:** (1) Certification & Traceability, (2) UK demand increase, (3) Productivity change, (4) Pandemic situation (COVID-19), (5) Government sustainability actions. Results compared across 1-year and 10-year horizons in a table.

- **Key finding:** Improving agricultural productivity is the most important factor for a sustainable coffee supply chain. Combining SD with multi-criteria decision making (TOPSIS) gives a more practical and accurate solution than either tool alone.

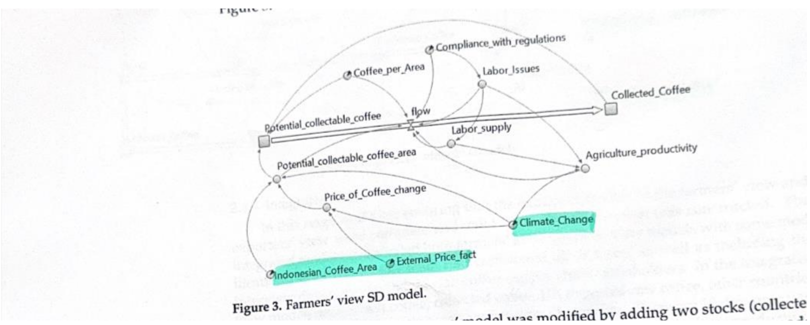



Figure 3. Farmers' view SD model.

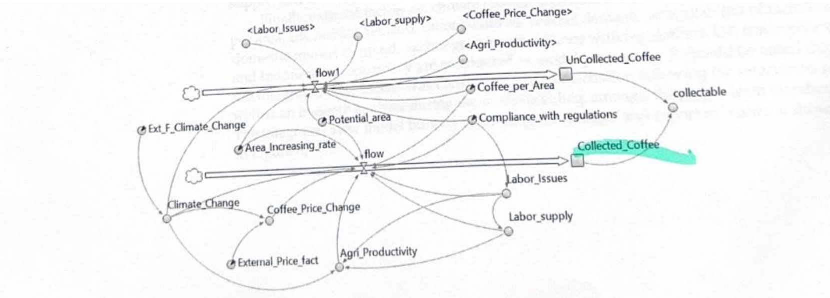
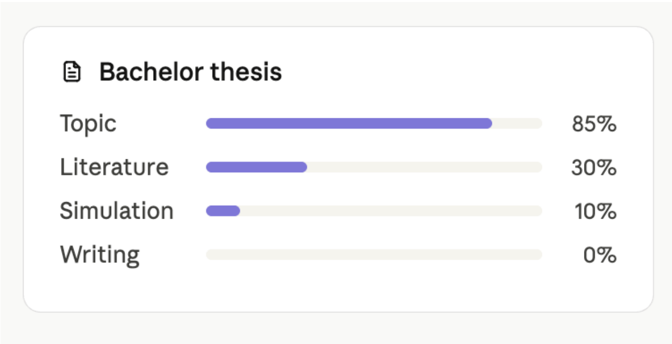



Figure 4. Revised Farmer' view SD model.




**Paper 5: Awudu & Zhang (2021) "A System Dynamics Approach to Comparative Analysis of Biomass Supply Chain Coordination Strategies"**

Scope: Forestry biomass supply chain - from forest sites through hubs to community energy conversion facilities. Models coordination strategies between suppliers, hubs, and energy consumers.

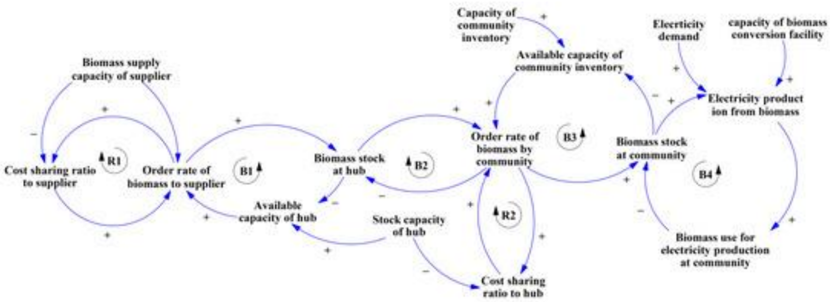
Method: System Dynamics simulation-based optimization. Uses stock-flow diagrams with full Equations.

Inputs: Biomass stock levels at suppliers/hubs/communities, yield levels and productivity at forestry sites, purchase costs, holding costs, delivery costs, bioenergy conversion costs, order quantities, price factors, cost-sharing ratios.

Outputs: Total supply chain cost minimization under two coordination strategies. Behavior-over-time for biomass stock levels, cumulative costs, and energy conversion accross the chain.

Model structure: Stock-flow diagram with stocks at three levels (supplier, hub, community) connected by inflows and outflows. 2 reinforcing loops (R1: quantity discount, R2: cost-sharing) and 3 balancing loops (B1-B3: storage capacity constraints). Compares two coordination scenarios: quantity discount strategy vs cost-sharing strategy.

Key finding: Coordination strategies significantly reduce total supply chain cost compared to no coordination. Cost-sharing outpreforms quantity discounts for smaller supply chains, while quantity discounts work better at scale.



Paper 6: Hartwick, Ismail, Novais, Zeeshan & Ehm (2023) "Operational Strategies for Managing Supply Chain Disruptions: Simulation of External Factor Impact on a Semiconductor Industry"

Scope: Semiconductor supply chain - five echelons from silicon supplier through manufacturer, Tier- 2, Tier-1 to OEM. Models how external disruptions (pandemics, extreme weather, geopolitical tensions) cascade through the chain.

Method: System Dynamics, implemented in AnyLogic. Uses historic global light vehicle sales as exogenous demand signal.

Inputs: End market demand (global light vehicle sales), order signals propagating upstream through 5 echelons, production capacities at each level, inventory coverage, lead times, disruption parameters for 4 risk types.

Outputs: Impact on customer orders, revenue, and distribution center stock under different disruption scenarios. Measures amplitude and duration of disruption impact at each echelon. Model structure: 5-echelon supply chain (silicon supplier, semiconductor manufacturer, Tier-2, Tier-1, OEM). Demand signal propagates upstream creating bullwhip effect. Tests 4 disruption types and evaluates operational strategies for each. Includes distribution center system dynamics structure.

Key finding: Upstream companies suffer the most from disruptions in both amplitude and duration. Restoring equillibrium after a disruption takes longer than expected. Close collaboration and trust between supply chain players - especially transparency in ordering behavior, inventory coverage, and lead time communication - are critical for resilience.